Project Progress Report: Modelling the Spread of Information Through Social Media Reposting

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Abstract

We model how a single piece of content spreads when users can be exposed repeatedly before reposting or losing interest. The formulation uses exposure classes $I_0, \ldots, I_m$, active sharers S, and saturated users R, with a mean-field exposure rate $\lambda = \beta S/N$. This progress report gives the full model specification, literature links, a preliminary threshold heuristic, partial equilibrium and stability discussion for $m = 2$, Python simulations (viral, moderate, and fizzle regimes), parameter tables, four figures (including a sensitivity curve), and a short roadmap for the remaining work.

1 Introduction

Social media platforms such as Instagram play a major role in how information spreads. Content gains visibility through reposting, recommendations, and repeated views; Meta describes how ranking combines signals such as engagement and recency (Meta Platforms Inc., 2024). Users often see the same post or reel more than once before reposting, so exposure is *cumulative*.

Many classical diffusion models assume a fixed exposure threshold or a single "infected" state. Real behaviour is heterogeneous: some users share quickly, others need several exposures, and some disengage. This project uses multiple exposure classes I_i so that sharing or saturation can occur at any level $i \in \{0, \ldots, m\}$, within a tractable ODE framework.

Research questions. (1) When does reposting amplify from a small seed? (2) How do β (repost intensity) and γ (attention decay) affect peaks and final reach? (3) Can the same equations represent qualitatively different content types (e.g. memes vs. news)?

2 Literature Review

Information diffusion is often modelled with epidemic-style compartments. Daley and Kendall (1965) introduced a rumor model with ignorants, spreaders, and stiflers; Maki and Thompson (1973) gave a related variant. Kempe, Kleinberg, and Tardos (2003) studied influence maximization on networks. Centola (2010) showed experimentally that repeated exposure can be necessary for adoption (*complex contagion*). Rogers (2003) described heterogeneous adoption timing across adopter categories. Granovetter (1978) used threshold models where decisions depend on accumulated social influence; our (ρ_i, σ_i) play an analogous role at each exposure stage.

This project extends rumor/SIR-style ideas with an explicit chain $I_0 \to I_1 \to \cdots \to I_m$, consistent with feed dynamics (Meta Platforms Inc., 2024).

3 Model Formulation and Justification

3.1 Compartments

Let N be fixed. For $i = 0, \ldots, m$, $I_i(t)$ counts users exposed exactly i times who have not started sharing; $S(t)$ counts active sharers; $R(t)$ counts saturated users. As summarized in **Table 1**,

$$\sum_{i=0}^{m} I_i(t) + S(t) + R(t) = N.$$

The exposure rate is $\lambda(t) = \beta S(t)/N$ with $\beta > 0$. Upon an exposure in I_i, a user shares with probability ρ_i, saturates with probability σ_i, or advances to I_{i+1} with probability $1 - \rho_i - \sigma_i$, where $0 \le \rho_i, \sigma_i \le 1$ and $\rho_i + \sigma_i \le 1$. Sharers decay at rate $\gamma > 0$.

3.2 Assumptions

1. Closed audience: N constant over the study horizon.
2. Homogeneous mixing: $\lambda = \beta S/N$.
3. Probabilistic local rules (ρ_i, σ_i) at each exposure.
4. Single piece of content (no competing posts).

3.3 Link to fixed-threshold models

If $\rho_i = \sigma_i = 0$ for all $i < m$, users cannot share or saturate until I_m; the next exposure forces a decision via ρ_m and σ_m. The general model therefore *strictly generalizes* a rigid "exactly m exposures then decide" rule.

3.4 Equations

$$\frac{dI_0}{dt} = -\lambda(t) I_0, \tag{1}$$

$$\frac{dI_i}{dt} = (1 - \rho_{i-1} - \sigma_{i-1})\lambda(t) I_{i-1} - \lambda(t) I_i, \quad i = 1, \ldots, m-1, \tag{2}$$

$$\frac{dI_m}{dt} = (1 - \rho_{m-1} - \sigma_{m-1})\lambda(t) I_{m-1} - (\rho_m + \sigma_m)\lambda(t) I_m, \tag{3}$$

$$\frac{dS}{dt} = \sum_{i=0}^{m} \rho_i \lambda(t) I_i - \gamma S, \tag{4}$$

$$\frac{dR}{dt} = \sum_{i=0}^{m} \sigma_i \lambda(t) I_i + \gamma S. \tag{5}$$

Interpretation. Equation (1) removes users from I_0 at rate λI_0. Equations (2)–(3) move mass along the exposure chain and allow exit to S or R from I_m. Equation (4) balances inflow $\sum_i \rho_i \lambda I_i$ and outflow γS; thus $dS/dt > 0$ when inflow exceeds outflow. Equation (5) collects saturation from exposures and decay from S.

3.5 Table 1: Model variables and parameters

Symbol	Description
N	Total population (fixed)
m	Maximum exposure index
$I_i(t)$	Exposed i times, not yet sharing
$S(t)$	Active sharers
$R(t)$	Saturated users
β	Repost intensity in $\lambda = \beta S/N$
γ	Sharer decay rate
ρ_i, σ_i	Share / saturate probabilities at exposure i

4 Preliminary Results

4.1 Threshold heuristic

Near a disease-free state ($I_0 \approx N$, $S \approx 0$), define

$$R_{\text{eff}} \approx \frac{\beta}{\gamma}\, p_{\text{share}}, \tag{6}$$

where p_{share} approximates the probability that a user starting in I_0 eventually enters S. For $m = 2$,

$$p_{\text{share}} \approx \rho_0 + (1 - \rho_0 - \sigma_0)\rho_1 + (1 - \rho_0 - \sigma_0)(1 - \rho_1 - \sigma_1)\rho_2. \tag{7}$$

Large β or ρ_i raises R_{eff}; large γ or σ_i lowers it. This scalar guides scenario design and matches the ordering of simulated peaks in §4.4.

4.2 Equilibria and partial stability ($m = 2$)

Disease-free equilibrium. One equilibrium is $E_0 = (I_0, I_1, I_2, S) = (N, 0, 0, 0)$, with $R = 0$ and $\lambda = 0$: no sharing and no exposures.

Linearization at E_0. With state (I_0, I_1, I_2, S) and $\lambda = \beta S/N$, the Jacobian at E_0 has the block form (derivatives evaluated at $I_0 = N$, $I_1 = I_2 = S = 0$, $\lambda = 0$):

$$J(E_0) = \begin{pmatrix} 0 & 0 & 0 & -\beta \\ 0 & 0 & 0 & (1 - \rho_0 - \sigma_0)\beta \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \rho_0\beta - \gamma \end{pmatrix}.$$

Eigenvalues include 0 (multiplicity three, reflecting the degeneracy of the linearization at $\lambda = 0$ along the exposure chain) and $\lambda_S = \rho_0\beta - \gamma$. The sign of λ_S governs the immediate response of S to a tiny seed when only the $I_0 \to S$ shortcut matters; *secondary* inflow into S through I_1, I_2 is not visible in this leading 4×4 block alone. That is why simulations are essential and why we use R_{eff} in (6) as a practical summary. A complete spectral analysis of the nonlinear system and characterization of endemic equilibria will be finished in the final write-up.

4.3 Computational setup

Simulations use Python 3 and `scipy.integrate.odeint` (`simulate_instagram_ode.py`). We set $m = 2$, $N = 1000$, initial condition $I_0(0) = 985$, $I_1(0) = I_2(0) = 0$, $S(0) = 15$, $R(0) = 0$, and horizon $t \in [0, 120]$ in **dimensionless time units** (common scale across scenarios). Table 2 gives outcomes; Table 3 gives (ρ_i, σ_i) and (β, γ).

Table 1: Simulation outcomes ($N = 1000$, seed $S(0) = 15$, $t \in [0, 120]$).

Scenario	β	γ	**Peak** $S(t)$	**Time to peak**	$R(120)$
Viral	1.50	0.080	321.66	51.29	981.56
Moderate	1.13	0.089	171.51	114.39	593.46
Fizzle	0.15	0.600	15.00	0.00	15.41

Table 2: Parameters by scenario ($m = 2$). Moderate uses the same (ρ_i, σ_i) as viral but lower β/γ.

Scenario	(β, γ)	(ρ_0, ρ_1, ρ_2), $(\sigma_0, \sigma_1, \sigma_2)$
Viral	(1.50, 0.080)	(0.05,0.15,0.25), (0.01,0.02,0.04)
Moderate	(1.13, 0.089)	(0.05,0.15,0.25), (0.01,0.02,0.04)
Fizzle	(0.15, 0.600)	(0.01,0.02,0.03), (0.10,0.15,0.20)

Using (7): viral and moderate share $p_{\text{share}} \approx 0.34$; fizzle has $p_{\text{share}} \approx 0.05$. The moderate scenario lowers β/γ enough to cut the peak roughly in half compared with viral, while fizzle leaves $S(t)$ at the seed.

4.4 Figures and sensitivity

Figure 1 contrasts viral and fizzle $S(t)$ and $R(t)$. Figure 2 adds the **moderate** curve: a clear middle peak between viral and fizzle. Figure 3 shows all compartments for the viral run. Figure 4 plots $\max_t S(t)$ versus β with $\gamma = 0.09$ and viral-shaped (ρ_i, σ_i): peak sharing increases sharply as β rises, supporting the role of repost intensity in our research questions.

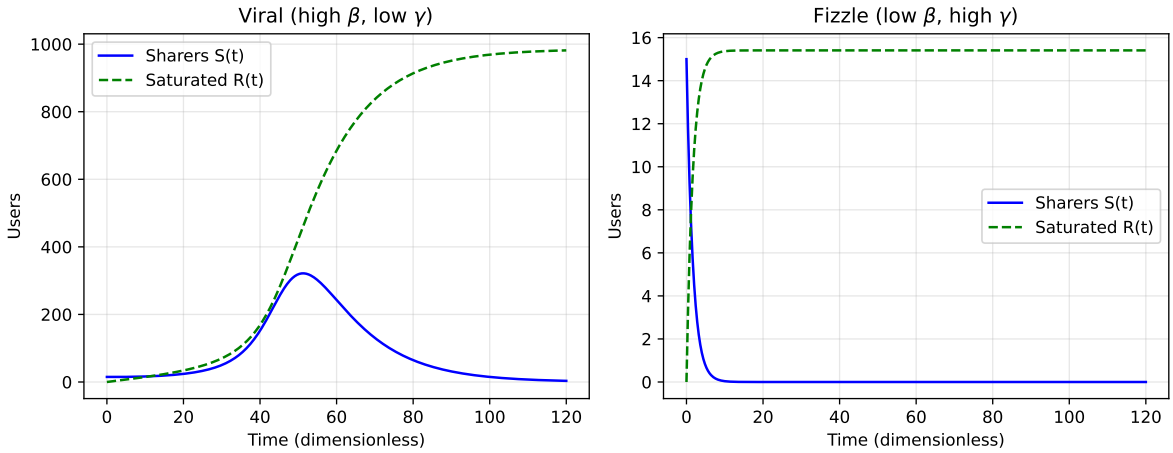


Figure 1: Viral (left) vs. fizzle (right): $S(t)$ and $R(t)$.

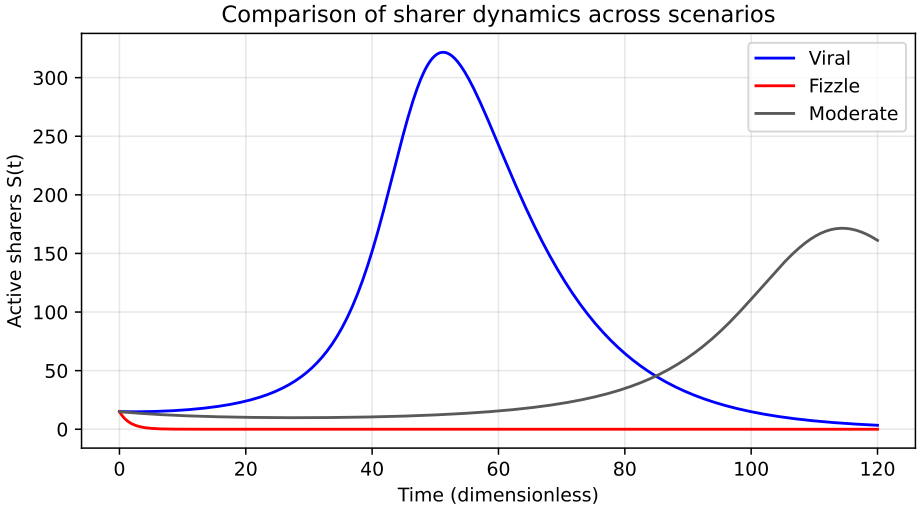


Figure 2: Active sharers $S(t)$: viral, moderate, and fizzle.

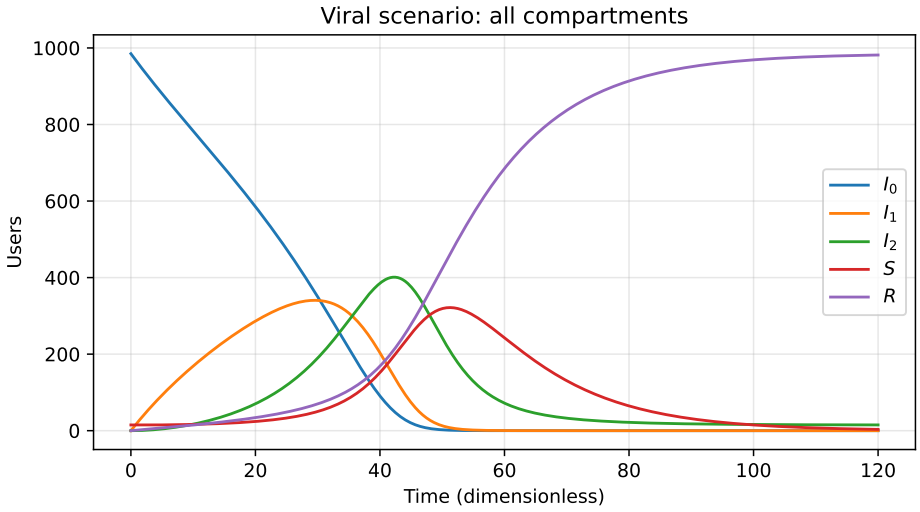


Figure 3: Viral scenario: I_0, I_1, I_2, S, R.

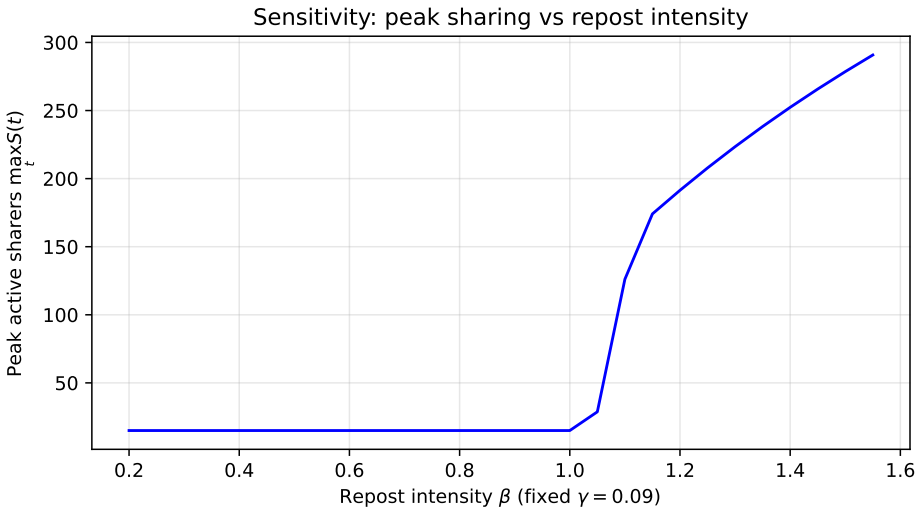


Figure 4: Sensitivity: peak $\max_t S(t)$ vs. β with fixed $\gamma = 0.09$ (viral (ρ_i, σ_i)).

4.5 Qualitative case-study mapping (illustrative)

- **Viral meme:** high β, low γ, and relatively large early ρ_i produce a fast, tall peak (Table 2, viral row).
- **Moderate spread:** same reinforcement pattern but reduced β/γ yields a slower, smaller cascade—useful for "breakout but not explosion" narratives.
- **Fizzle / low engagement:** low β, high γ, and large σ_i shut down growth even with a seed (fizzle row).

These mappings are interpretive, not calibrated to data.

4.6 Discussion of progress and limitations

Strengths. Clear structure for the rubric: problem motivation, literature, justified model with Table 1 referenced in text, analytical partial results, and rich numerical results (four figures, two tables) with interpretation.

Limitations. Mean-field mixing; parameters illustrative; Jacobian at E_0 is structurally degenerate ($\lambda = 0$), so full stability requires careful treatment of the nonlinear chain (in progress).

5 Future Work

1. Complete eigenvalue/stability analysis for E_0 and characterize positive equilibria (if any) for $m = 2$.
2. Extend sensitivity: grids over γ and (ρ_0, ρ_1, ρ_2); optional heatmaps.
3. Optional: degree-weighted exposure or a small network toy model (Kempe et al., 2003).
4. Final polish: notation pass, tie one scenario loosely to a real campaign or dataset if feasible.

6 References

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